

Assignment

Objective

The Objective is to train a Segmentation model using a Mobile Net pretrained on the Imagenet Dataset as an encoder and design a decoder that predicts Segmented masks

- Decoder should be constructed in this Assignment. We need to do the feature extraction and the fine tuning
- Coming to Feature extraction. Freeze the encoder only the Decoder is trained
- Coming to Fine tuning. Last layer of encoder along with Decoder get trained.
- The Dataset need to use here is ISIC 2016
- And Calculate the Intersection over Union and Dice score.

Preprocessing the Dataset

- We have the Original Skin ^{Image} ISIC and Segmented mask
- Convert the Image to $(128, 128)$ Tensor. and also for the Segmented mask $(128, 128)$ Tensor
- Image is RGB and mask is Gray scale
- Then pass the Image tensor and the mask tensor to the train loader.
- mask is converted to binary. black & white

- Here we are using the Unit Architecture

- Mobile net V2 encoder

- Our decoder But Concatenating Skip Connections

- In the code we broke the encoder into 5 layers

$$[B, 3, 128, 128] \xrightarrow{\text{layer 1}} [B, 16, 64, 64]$$

$$[B, 16, 64, 64] \xrightarrow{\text{layer 2}} [B, 24, 32, 32]$$

$$[B, 24, 32, 32] \xrightarrow{\text{layer 3}} [B, 32, 16, 16]$$

$$[B, 32, 16, 16] \xrightarrow{\text{layer 4}} [B, 96, 8, 8]$$

$$[B, 96, 8, 8] \xrightarrow{\text{layer 5}} [B, 1280, 4, 4]$$

- Now coming to the decoder And it is of 4 blocks

$$[B, 1280, 4, 4] \xrightarrow{\text{block 1}} [B, 256, 8, 8]$$

$$[B, 256, 8, 8] \xrightarrow{\text{block 2}} [B, 128, 16, 16]$$

$$[B, 128, 16, 16] \xrightarrow{\text{block 3}} [B, 64, 32, 32]$$

$$[B, 64, 32, 32] \xrightarrow{\text{block 4}} [B, 32, 64, 64]$$

$$[B, 32, 64, 64] \xrightarrow{\text{final convolution}} [B, 1, 64, 64]$$

$$[B, 1, 64, 64] \xrightarrow{\text{interpolate}} [B, 1, 128, 128]$$

- finally Sigmoid Applied

Dice Loss

- Segmentation datasets are imbalanced

Background pixels $\gg$ lesion pixels

Dice formula

$$\frac{2(A \cap B)}{|A| + |B|}$$

where $A = \text{prediction}$
 $B = \text{ground truth}$

- intersection = (pred * target) . sum()

Both are overlapped But Both are 1 then 1 else 0

$$1 - \frac{2(A \cap B)}{|A| + |B|} \quad \text{this is Dice loss}$$

IoU

$$\text{IoU} = \frac{(A \cap B)}{(A \cup B)}$$

For fine tuning

- We unfreeze the layer 5 to train the layer 5
we use Adam optimizer $lr = 1e-5$

- to train the decoder $lr = 1e-4$

Parameters

batch size = 16

epochs = 20

	IOU score	Dice score
Feature extraction	0.83	0.90
Fine tuning	0.83	0.91